NOMURA

JPY Weekly Flow Monitor

Global Markets Research
16 August 2024

Foreign Exchange - Global

How can we estimate the size of yen carry trade positions from existing statistics?

A weekly update on key JPY flow developments.

Key points of the week

- *Although we judge it is not feasible to know the true size of yen-carry trade positions all over the world, we can track some proxies based on existing statistics.*
- *To estimate the size of on-balance yen-carry trade positions by offshore investors, we can use the BOJ's monthly data and the BIS's quarterly data, though there are some caveats. As for off-balance transactions, the well-known weekly positioning data released by the CFTC remain a good proxy, in our view.*
- *Given the frequency of data, at least to forecast short-term movement of foreign exchange rates, we believe the CFTC's positioning data is the most practical proxy for the size of yen-carry trading positions among existing statistics.*

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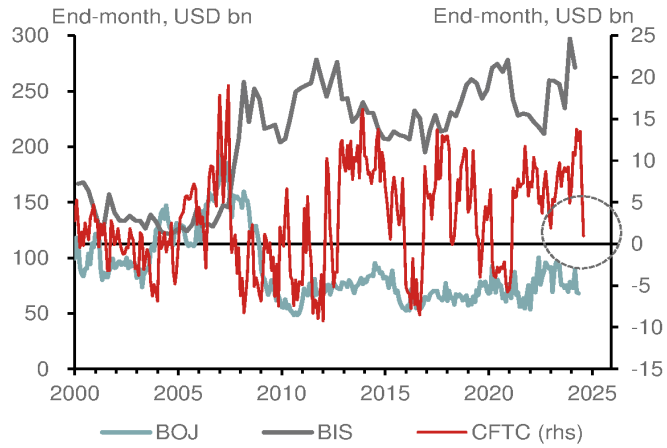
Fig. 1: Proxies for the size of yen-carry trading positions by offshore investors

Source	Data	Coverage				Latest data	Latest value ($ bn)	Frequency
		On-balance		Off-balance				
		Yen-carry	Others	Yen-carry	Others			
BOJ	Inter-office lending by Japan branches of foreign banks	✓	✓			31-May	68	Monthly
BIS	Yen-denominated claims to non-residents by banks	✓	✓			31-Mar	271	Quarterly
CFTC	Net short positions in currency futures by non-commercial traders			✓	✓	6-Aug	1	Weekly

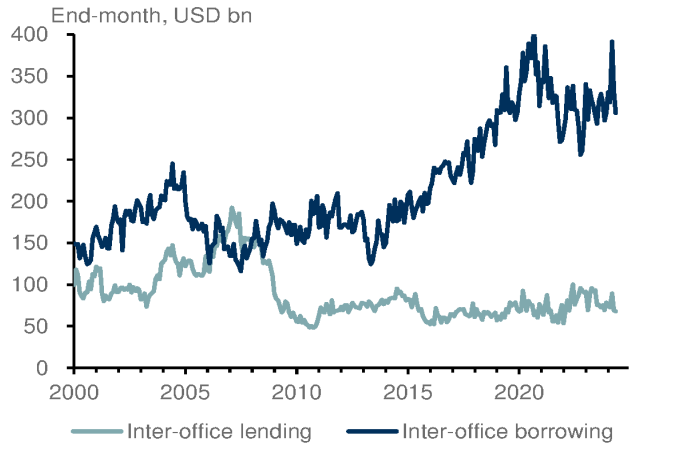
Note: BIS's yen-denominated claims are to non-banking sectors.
Source: BIS, BOJ, CFTC, Macrobond, Nomura

Fig. 2: Proxies to estimate the size of yen-carry trade positions



Note: "CFTC" is the net JPY short position of non-commercial traders.
Source: BIS, BOJ, Macrobond, Nomura

Fig. 3: Inter-office transactions by Japan branches of foreign banks



Source: Macrobond, Nomura

How can we estimate the size of yen carry trade positions from existing statistics?

Definition of yen-carry trade

As the yen's recent sharp appreciation from late July to early August is likely to have been caused mainly by *the large scale unwinding of yen-short positions, mainly for carry purposes*, markets seem to be wondering how much yen-carry trade positions are left and how we can know the size of them. Needless to say, it is not feasible to know the true size of yen-carry trade positions all over the world, especially in a timely manner, but we may

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See Appendix A-1 for analyst certification, important disclosures and the status of non-US analysts.

be able to estimate it from existing statistics.

Yen-carry trade is defined as making JPY short positions to mainly benefit from the interest rate gap between Japan and other countries such as the US. If the yen depreciates, or appreciates less than the return on the interest rate gap, investors can receive total positive returns from the position. Yen-carry trades are divided into two subcategories in how they appear on banks' (or investors') balance sheets.

1. **On-balance trades:** This is a simple and intuitive case of a yen-carry trade. Investors borrow the yen from banks and sell it in FX spot markets to buy higher-yielding assets, then buy back the yen to close the positions. This on-balance yen-carry trade positions appear as the lending of JPY in the banks' balance sheets. That said, banks may be less willing to be involved in this type of trade, because it has become more costly for banks to expand their balance sheets due to the introduction of international banking regulations such as leverage ratio after the Global Financial Crisis (GFC).
2. **Off-balance trades:** Investors can make yen-carry trade positions without borrowing yen from banks. For example, the spot rate of USD/JPY is now around 147, and the 3-month forward rate is around 145. The gap between the spot rate and the forward rate is basically decided by the interest rate gap between Japan and US. If investors now buy USD/JPY future (3-month) and sell spot USD/JPY in 3 months, they can receive the same return as the on-balance trades above without borrowing the yen (transactions costs are ignored), if the spot rate of USD/JPY is unchanged at 147 in 3 months.

Proxies for on-balance yen-carry trade positions

There are two proxies for the size of on-balance yen-carry trade positions (*Fig. 1*).

First, we often monitor the inter-office lending by Japan branches of foreign (non-Japanese) banks to their headquarters and other branches released by the BOJ. When offshore investors borrow the yen from the banks, foreign banks fund the yen via their Japan branches. Hence, we can presume the inter-office lending by Japan branches of foreign banks reflects the size of yen-carry trade positions indirectly. However, as the indicator also includes other banking flows, there is a risk of both overestimation and underestimation. For example, this indicator remained far below its pre-GFC peak as of end-May but picked up somewhat during 2022-2023 (*Fig. 2*). This may be attributable to the increase in inter-office borrowing during the late 2010s (*Fig. 3*). The aggregated inter-office lending shown in the figures above includes only individual branches that lend the yen to their headquarters and other branches on a net basis. Hence, the increase in inter-office borrowing (likely due to the increase in cross-currency swaps) might have limited the growth of the aggregated inter-office lending, even if the size of yen-carry trade positions might have increased during the same period.

Another proxy is global bank's yen-denominated cross-border lending to the offshore residents of non-banking sectors released by the BIS. This indicator was above its pre-GFC peak at end-March (*Fig. 2*). Although this indicator theoretically can track all offshore yen lending by banks, including on-balance yen-carry trades, it also likely includes other yen-denominated lending such as cross-border M&A-related loans. Hence, this indicator should be interpreted as the maximum of the estimated size of yen-carry trades, as Hyun Song Shin, the chief economist of the BIS, pointed out recently in a posting to the X platform. Moreover, this indicator is unsuitable for real-time tracking, as it is updated only quarterly, and we need to wait for the next data as of end-June to be released on 16 September. To gauge the magnitude of the recent position unwinding, we need to wait for December update.

While these two datasets may not capture the entire on-balance yen-carry trade positions, it is worth noting that the current peak is not as significant as the peak before the GFC, if we consider such factors as the expansion in the size of the economy, and FX market transaction volume..

Proxies for off-balance yen-carry trade positions

As a proxy for off-balance yen-carry trade positions, we suggest the well-known positioning data in currency futures of non-commercial traders released by the CFTC weekly (*Fig. 1*). By definition of the CFTC, the positions by non-commercial traders are aimed mainly at risk-taking rather than hedging; thus, we can presume the data track well the development of off-balance yen-carry trade positions, while positions may be not only based on carry purposes but on directional or fundamental views. It also should be also

noted that the data include only the positions of investors who report to the CFTC.

Given the frequency of data, at least to forecast short-term movement of foreign exchange rates, we believe the CFTC's positioning data are the most practical proxy for the size of yen-carry trade positions among existing statistics (*Fig. 1*). According to the CFTC's data, net JPY short positions were almost zero as of 6 August; thus, we believe most of yen-carry trade positions accumulated since the beginning of this year have already largely been unwound, which is likely to *decrease the likelihood of the yen's sharp appreciation in the near term*.

MOF international weekly capital flow data (4 to 10 August)

Japanese investor flows

- ***Equity***: Japanese investors net sold JPY328bn ($2.2bn) of foreign equities. With global stock prices declining sharply, Japanese investors, including investment trusts, seem to have net sold foreign equities in the week. Our high-frequency data indicate that net inflows into foreign equity investment trusts recorded *daily net outflows* for three consecutive business days (6-8 August) for the first time since last December (*Fig. 29*). Moreover, the relatively subdued gross purchase of foreign equities implies that pension funds had not yet begun buying foreign equities in earnest *to rebalance their portfolio allocations* as of last week.
- ***Long-term debt***: Japanese investors net purchased JPY1.5trn ($10.5bn) of foreign bonds, the largest amount since the second week of May. The global risk-off during the week brought about 'flight to quality', which encouraged investors to shift their funds from equities to bonds. Amid the global bond rally, Japanese investors seem to have net purchased safe-haven assets such as US Treasuries.

Foreign investor flows

- ***Equity***: Foreign investors net purchased JPY522bn ($3.6bn) of Japanese equities despite the global risk-off. Although they might have sold Japanese equities early last week, *the less hawkish speech by BOJ Deputy Governor Uchida on 7 August* reassured markets, which seem to have encouraged foreign investors to buy back Japanese equities late last week.
- ***Long-term debt***: Foreign investors net purchased JPY1.4trn ($9.8bn) of Japanese bonds, the largest amount since the first week of May. On 7 August, BOJ Deputy Governor, during his speech, clarified that the BOJ will not raise its policy rate when markets are unstable. In addition to the global bond rally discussed above, his less hawkish words made investors more confident that the upside potential of long-term JGB yields would be limited in the near term, which appears to have led to the large amount of net purchase of JGBs by foreign investors.

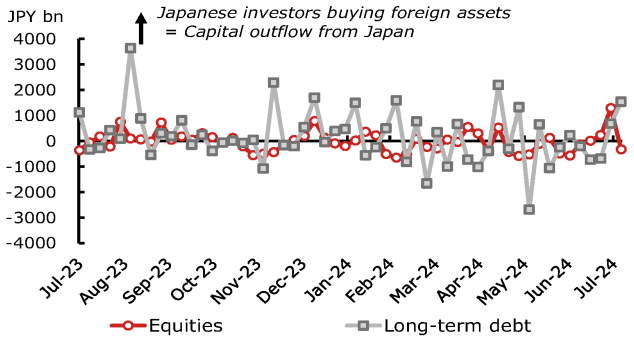
Fig. 4: Weekly MOF international transactions in securities (as of 4 – 10 August)

		Latest			Past 12W average		Past 12 weeks data
		JPY bn	$ bn	Z-score	JPY bn	$ bn	
Japanese investors	Foreign equities	-328	-2.2	-0.7	-128	-0.9	
	Foreign LT debt	1,539	10.5	1.5	-128	-0.9	
	Foreign ST debt	453.5	3.1	4.0	-30	-0.2	
Foreign investors	Japanese equities	522	3.6	0.8	20	0.1	
	Japanese LT debt	1,435	9.8	1.3	-29	-0.2	
	Japanese ST debt	562	3.8	0.3	-872	-5.9	

Note: 1. Positive signs in Japanese investors' foreign securities investment could indicate that they net bought foreign securities or capital outflow from Japan. 2. Positive signs in foreign investors' Japanese securities investment may indicate that they net bought Japanese securities or capital inflow to Japan. 3. Z-scores are calculated based on the past five years' data.
Source: Nomura, MOF

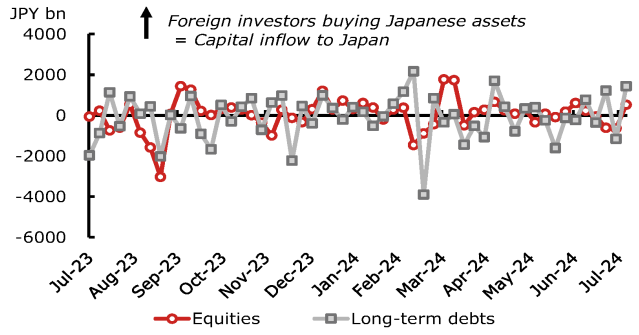
Weekly portfolio investment flow key charts

Fig. 5: Japanese investors' net purchases of foreign securities (weekly)



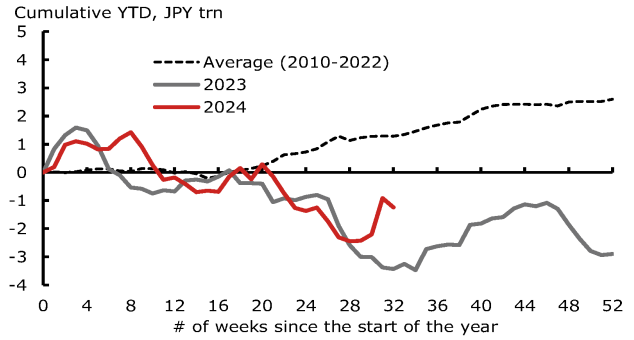
Source: Nomura, MOF

Fig. 6: Foreign investors' net purchases of Japanese securities (weekly)



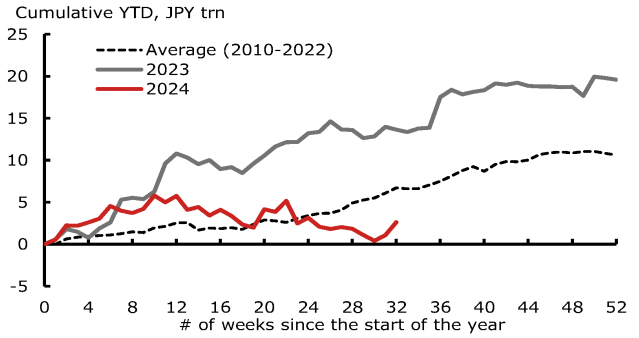
Source: Nomura, MOF

Fig. 7: Japanese investors' YTD cumulative net purchase of foreign equities



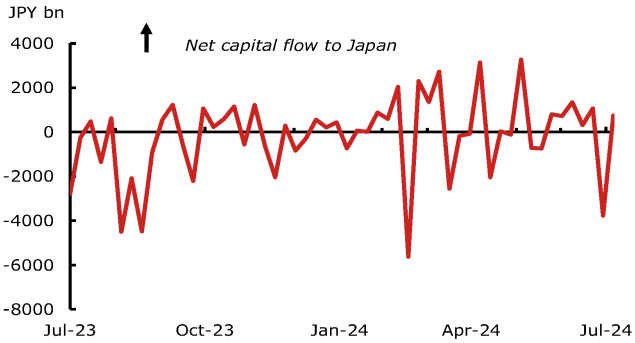
Source: Nomura, MOF

Fig. 8: Japanese investors' YTD cumulative net purchase of foreign bonds



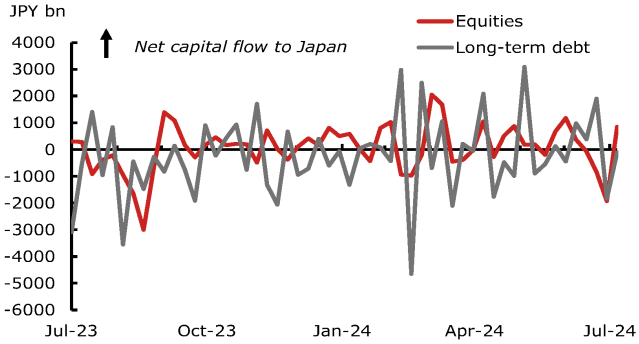
Source: Nomura, MOF

Fig. 9: Net capital flow to Japan via portfolio investment



Note: Sum of the portfolio investment assets and liabilities, with a positive sign meaning capital inflow to Japan.
Source: Nomura, MOF

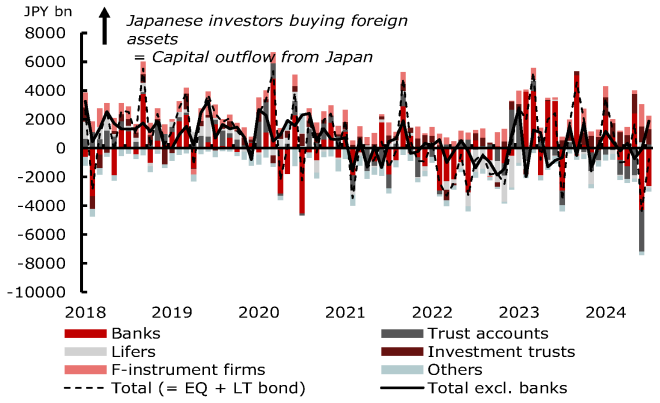
Fig. 10: Net capital flow to Japan by product



Note: Sum of the portfolio investment assets and liabilities, with a positive sign meaning capital inflow to Japan.
Source: Nomura, MOF

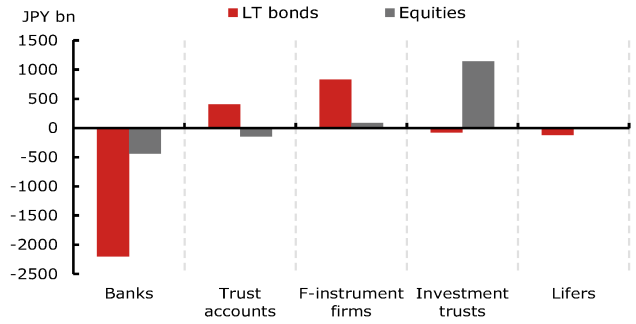
Monthly capital flow key charts

Fig. 11: Japan's net purchases of foreign securities by investor type



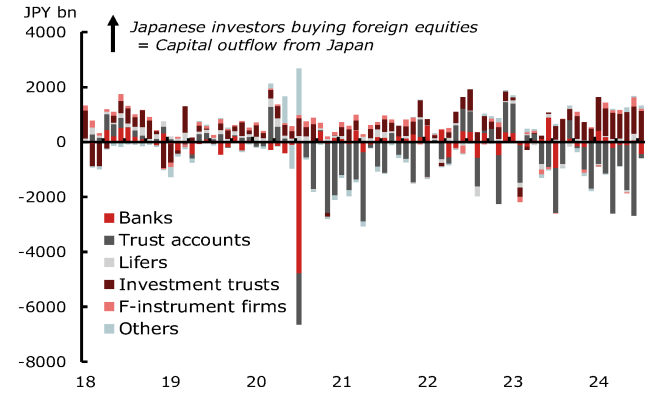
Source: Nomura, MOF

Fig. 12: Foreign portfolio investment by investor type



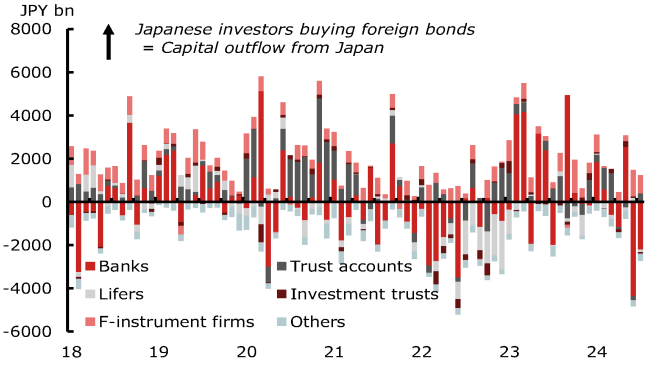
Note: Flow data for July 2024.
Source: Nomura, MOF

Fig. 13: Japan's net purchases of foreign equities



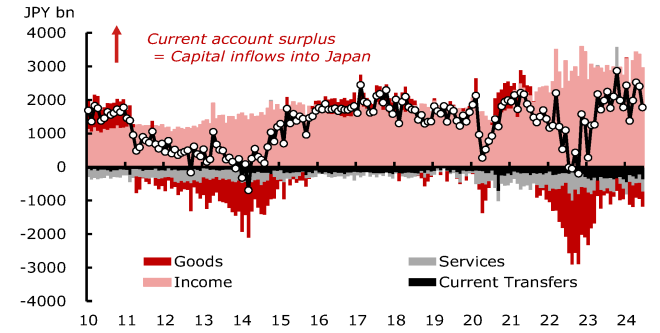
Source: Nomura, MOF

Fig. 14: Japan's net purchases of foreign bonds



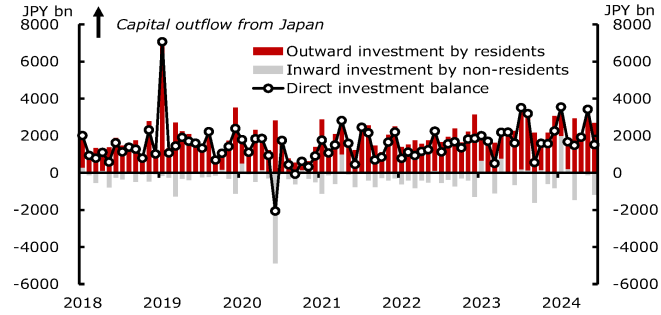
Source: Nomura, MOF

Fig. 15: Current account balance (SA)



Source: Nomura, MOF, BOJ

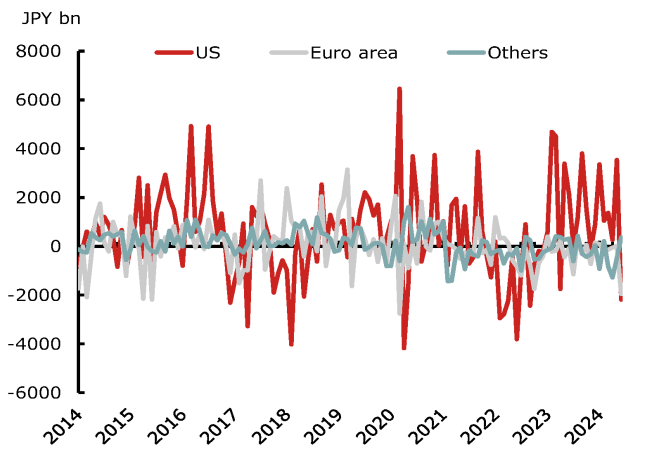
Fig. 16: Japan's direct investment balance



Note: Monthly data.
Source: Nomura, MOF, BOJ

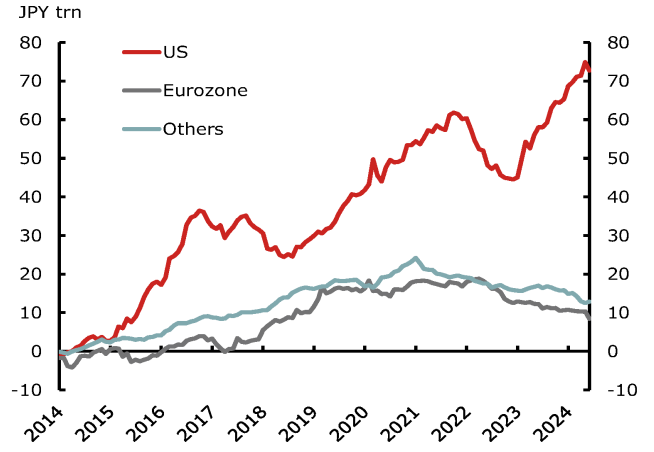
Portfolio investment flow by country

Fig. 17: Net purchases of foreign LT debt



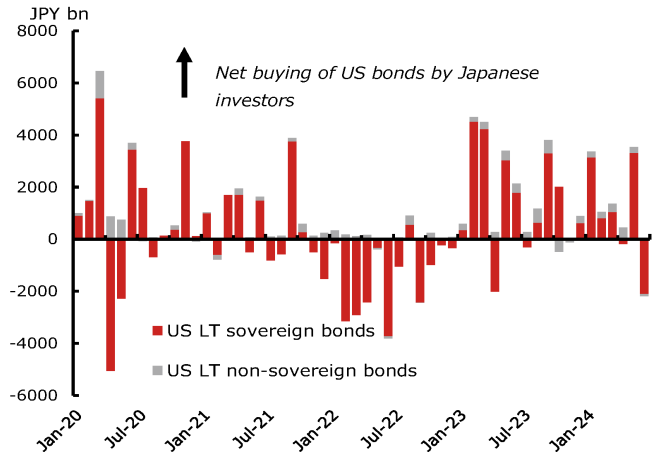
Source: Nomura, MOF, BOJ

Fig. 18: Cumulative net purchases of foreign LT debt



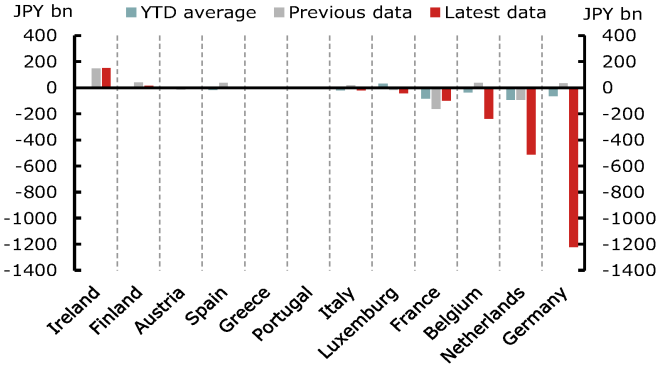
Note: Cumulative net purchase since 2014.
Source: Nomura, MOF, BOJ

Fig. 19: Japanese investors' US bond investment



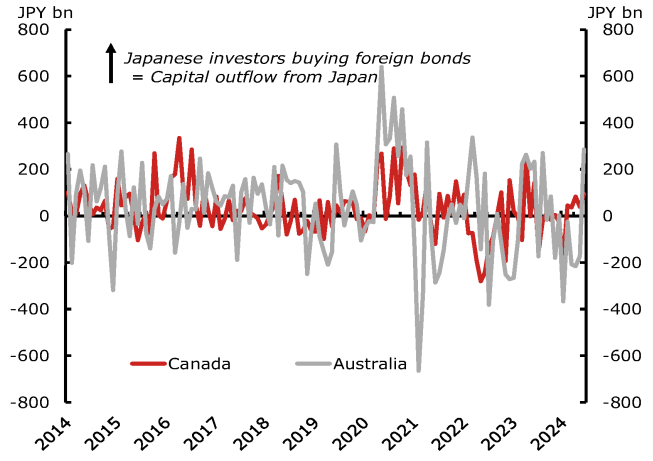
Source: Nomura, MOF, BOJ

Fig. 20: Japanese investors' euro area bond investment



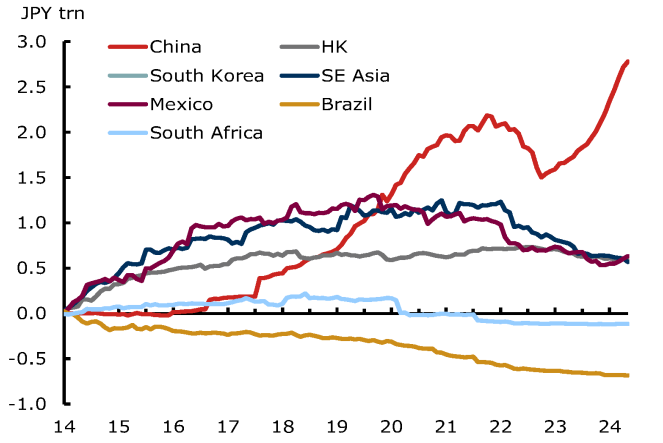
Note: The latest data is for June 2024.
Source: Nomura, MOF, BOJ

Fig. 21: Japanese investors' Australia and Canada bond investment



Source: Nomura, MOF, BOJ

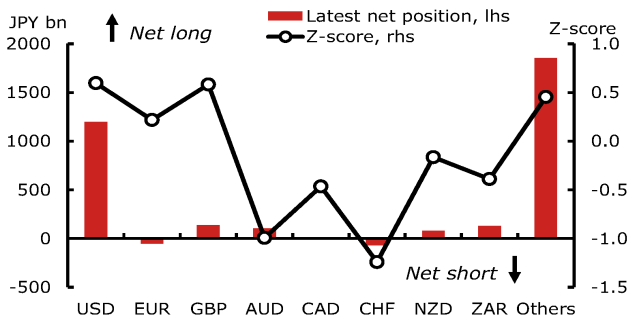
Fig. 22: Cumulative net purchase of EM countries' LT debt



Note: Cumulative net purchase since January 2014.
Source: Nomura, MOF, BOJ

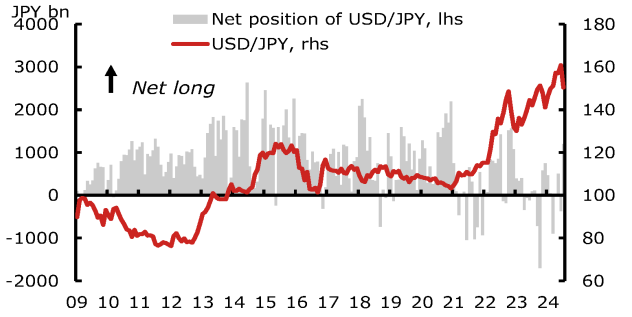
Margin trade positions

Fig. 23: Margin trade positions at OTC



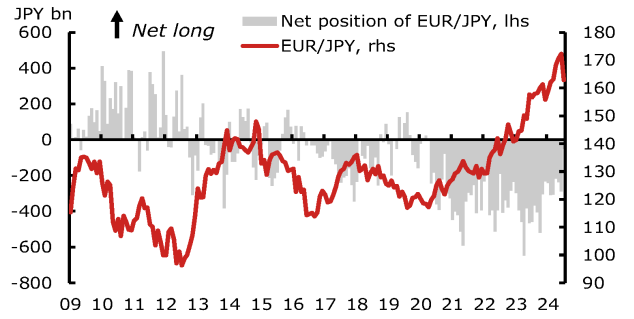
Note: Z-scores are calculated based on data since 2010. The Latest data as of July-2024.
Source: Nomura, Financial Futures Association of Japan, Macrobond

Fig. 24: USDJPY and margin trade positions at OTC



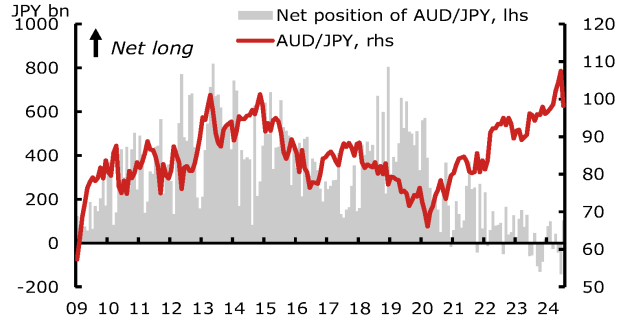
Source: Nomura, Financial Futures Association of Japan, Bloomberg

Fig. 25: EUR/JPY and margin trade positions at OTC



Source: Nomura, Financial Futures Association of Japan, Bloomberg

Fig. 26: AUD/JPY and margin trade positions at OTC



Source: Nomura, Financial Futures Association of Japan, Bloomberg

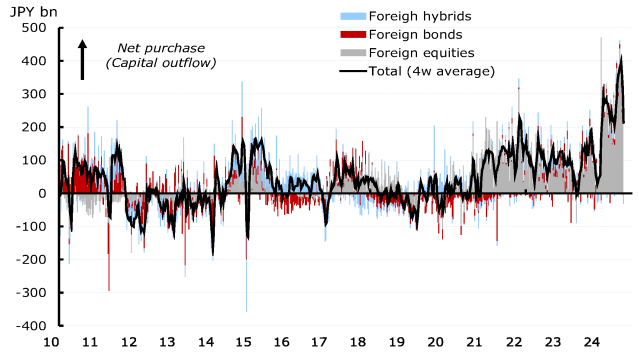
Fig. 27: Margin trading positioning at TFX

		USDJPY	EURJPY	GBPJPY	AUDJPY	NZDJPY	CADJPY	ZARJPY	TRYJPY	MXNJPY	Total
2024/8/13 (Latest)	JPYbn	-356.98	-14.83	-104.81	-0.53	0.32	-3.37	31.27	28.13	51.56	-369.24
	USDbn	-2.43	-0.10	-0.71	-0.004	0.002	-0.023	0.21	0.19	0.35	-2.51
2024/8/6	JPYbn	-355.67	-18.32	-101.65	-0.45	0.62	-4.64	31.19	27.24	45.42	-376.26
	USDbn	-2.46	-0.13	-0.70	-0.003	0.004	-0.032	0.22	0.19	0.31	-2.61
2024/7/30	JPYbn	-371.10	-22.32	-107.50	0.15	6.16	1.21	36.80	38.08	121.50	-297.03
	USDbn	-2.43	-0.15	-0.70	0.001	0.040	0.008	0.24	0.25	0.80	-1.94
2024/7/23	JPYbn	-372.16	-24.01	-110.58	-0.93	7.68	1.74	38.05	38.78	141.94	-279.48
	USDbn	-2.39	-0.15	-0.71	-0.006	0.049	0.011	0.24	0.25	0.91	-1.80

Source: Bloomberg, Financial Futures Association of Japan, TFX, Nomura

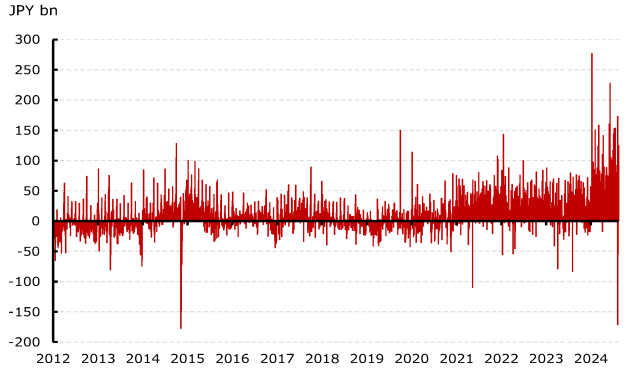
Toshin market

Fig. 28: Net purchases of toshins investing in foreign securities (weekly)



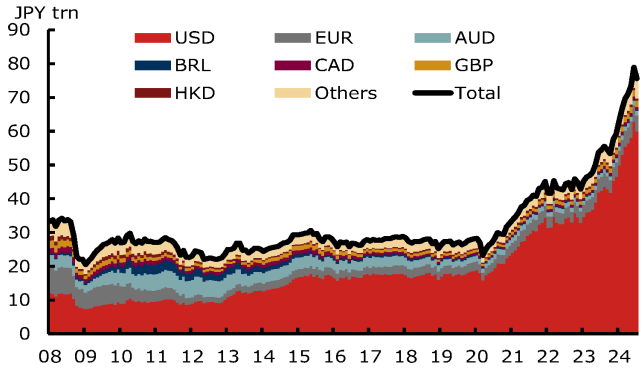
Source: NRI, Nomura

Fig. 29: Net purchases of toshins investing in foreign securities (daily)



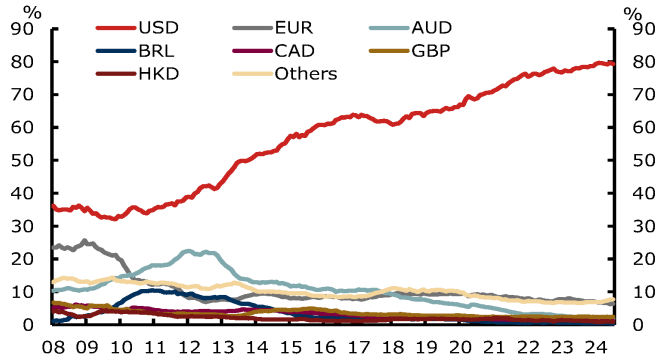
Source: NRI, Nomura

Fig. 30: Outstanding of toshins investing in foreign assets (classified by underlying assets)



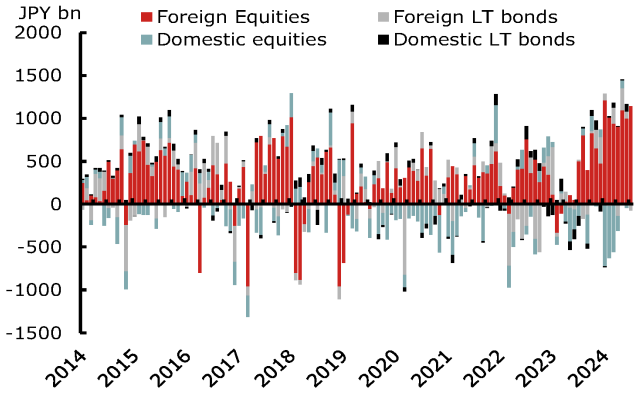
Source: Investment Trust Association, Japan, Nomura

Fig. 31: Share of outstanding toshins investing in foreign assets



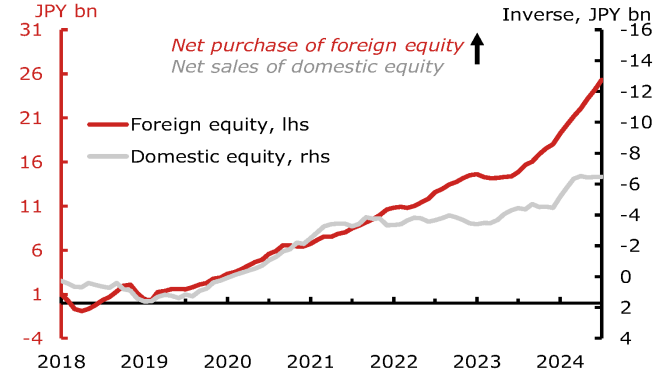
Source: Investment Trust Association Japan, Nomura

Fig. 32: Portfolio investment flow via investment trust



Source: Nomura, MOF, JSDA, JPX

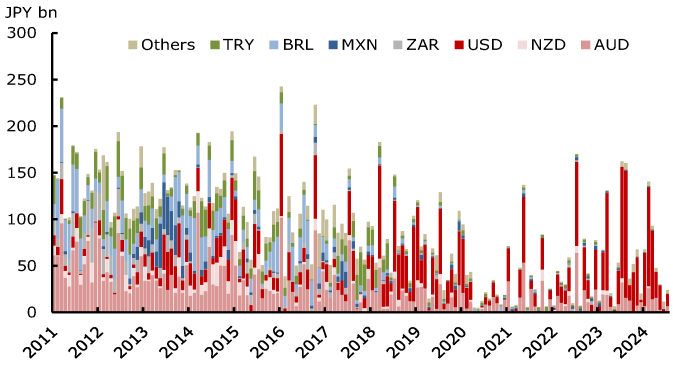
Fig. 33: Cumulative equity investment by toshins



Note: Aggregate of flow data since January 2018.
Source: Nomura, MOF, JPX

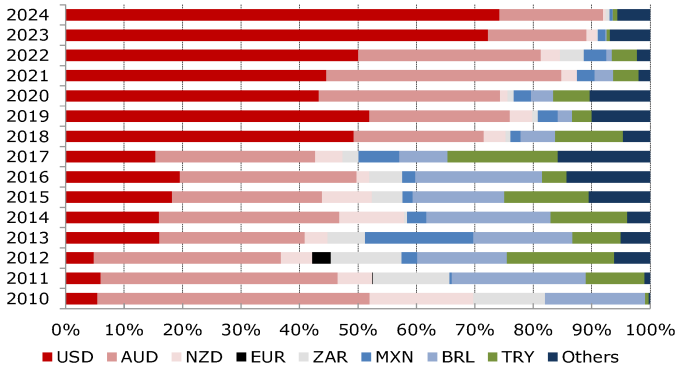
Uridashi bond market

Fig. 34: Uridashi issuance by currency



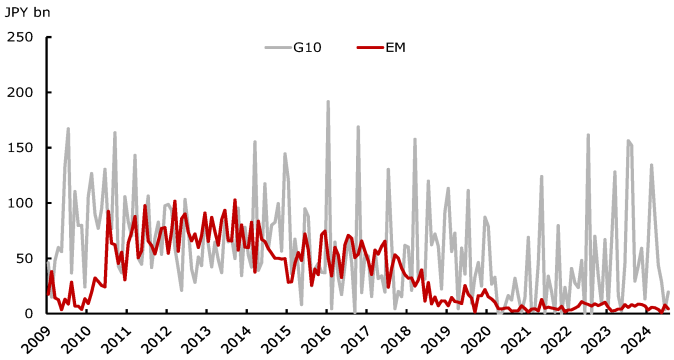
Source: Nomura, Bloomberg

Fig. 35: Major currency share in Uridashi issuance



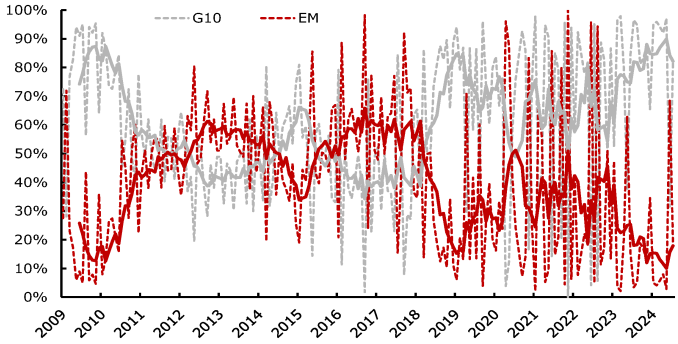
Note: Data for 2024 is YTD.
Source: Nomura, Bloomberg

Fig. 36: Uridashi bond issuance by type of currency



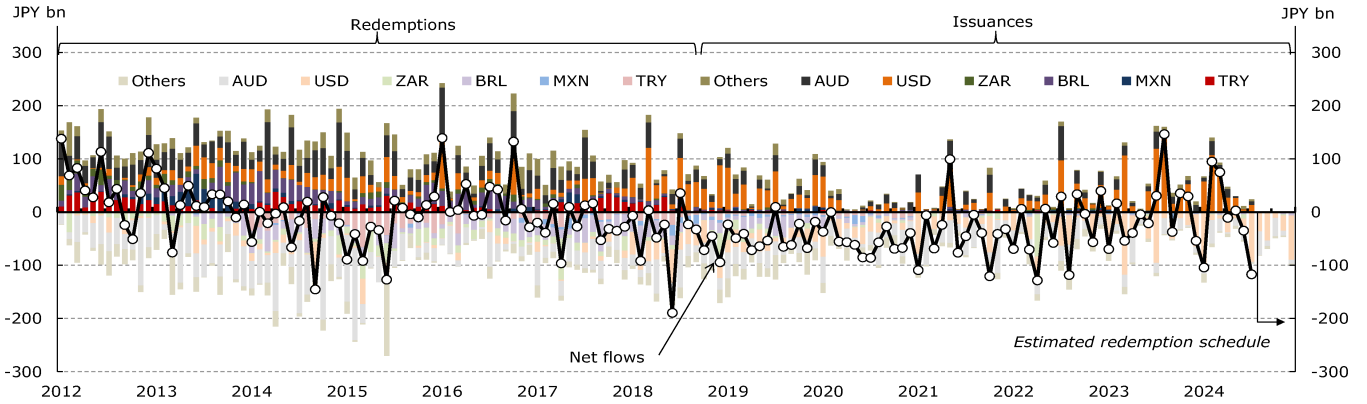
Source: Nomura, Bloomberg

Fig. 37: Share of Uridashi issuance by currency type



Source: Nomura, Bloomberg

Fig. 38: Uridashi bond issuance and redemptions by currency



Source: Nomura, Bloomberg

Appendix A-1

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• Quantitative analysis of price variations.
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Transactions involving convertible bonds are subject to a sales commission of up to 1.10% (tax included) of the transaction amount (or a commission of ¥4,400 (tax included) if this would be less than ¥4,400). When convertible bonds are purchased via OTC transactions (including offerings), only the purchase price shall be paid, with no sales commission charged. However, Nomura Securities may charge a separate fee for OTC transactions, as agreed with the customer. Convertible bonds carry the risk of losses owing to factors such as interest rate fluctuations and price fluctuations in the underlying stock. In addition, convertible bonds denominated in foreign currencies also carry the risk of losses owing to factors such as foreign exchange rate fluctuations.
When bonds are purchased via public offerings, secondary distributions, or other OTC transactions with Nomura Securities, only the purchase price shall be paid, with no sales commission charged. Bonds carry the risk of losses, as prices fluctuate in line with changes in market interest rates. Bond prices may also fall below the invested principal as a result of such factors as changes in the management and financial circumstances of the issuer, or changes in third-party valuations of the bond in question. In addition, foreign currency-denominated bonds also carry the risk of losses owing to factors such as foreign exchange rate fluctuations.
When Japanese government bonds (JGBs) for individual investors are purchased via public offerings, only the purchase price shall be paid, with no sales commission charged. As a rule, JGBs for individual investors may not be sold in the first 12 months after issuance. When JGBs for individual investors are sold before maturity, an amount calculated via the following formula will be subtracted from the par value of the bond plus accrued interest: (1) for 10-year variable rate bonds, an amount equal to the two preceding coupon payments (before tax) x 0.79685 will be used, (2) for 5-year and 3-year fixed rate bonds, an amount equal to the two preceding coupon payments (before tax) x 0.79685 will be used.
When inflation-indexed JGBs are purchased via public offerings, secondary distributions (uridashi deals), or other OTC transactions with Nomura Securities, only the purchase price shall be paid, with no sales commission charged. Inflation-indexed JGBs carry the risk of losses, as prices fluctuate in line with changes in market interest rates and fluctuations in the nationwide consumer price index. The notional principal of inflation-indexed JGBs changes in line with the rate of change in nationwide CPI inflation from the time of its issuance. The amount of the coupon payment is calculated by multiplying the coupon rate by the notional principal at the time of payment. The maturity value is the amount of the notional principal when the issue becomes due. For JI17 and subsequent issues, the maturity value shall not undercut the face amount.
Purchases of investment trusts (and sales of some investment trusts) are subject to a purchase or sales fee of up to 5.5% (tax included) of the transaction amount. Also, a direct cost that may be incurred when selling investment trusts is a fee of up to 2.0% of the unit price at the time of redemption. Indirect costs that may be incurred during the course of holding investment trusts include, for domestic investment trusts, an asset management fee (trust fee) of up to 5.5% (tax included/annualized basis) of the net assets in trust, as well as fees based on investment performance. Other indirect costs may also be incurred. For foreign investment trusts, indirect fees may be incurred during the course of holding such as investment company compensation.
Investment trusts invest mainly in securities such as Japanese and foreign equities and bonds, whose prices fluctuate. Investment trust unit prices fluctuate owing to price fluctuations in the underlying assets and to foreign exchange rate fluctuations. As such, investment trusts carry the risk of losses. Fees and risks vary by investment trust. Maximum applicable fees are subject to change; please thoroughly read the written materials provided, such as prospectuses or documents delivered before making a contract.
In interest rate swap transactions and USD/JPY basis swap transactions ("interest rate swap transactions, etc."), only the agreed transaction payments shall be made on the settlement dates. Some interest rate swap transactions, etc. may require pledging of margin collateral. In some of these cases, transaction payments may exceed the amount of collateral. There shall be no advance notification of required collateral value or collateral ratios as they vary depending on the transaction. Interest rate swap transactions, etc. carry the risk of losses owing to fluctuations in market prices in the interest rate, currency and other markets, as well as reference indices. Losses incurred as such may exceed the value of margin collateral, in which case margin calls may be triggered. In the event that both parties agree to enter a replacement (or termination) transaction, the interest rates received (paid) under the new arrangement may differ from those in the original arrangement, even if terms other than the interest rates are identical to those in the original transaction. Risks vary by transaction. Please thoroughly read the written materials provided, such as documents delivered before making a contract and disclosure statements.
In OTC transactions of credit default swaps (CDS), no sales commission will be charged. When entering into CDS transactions, the protection buyer will be required to pledge or entrust an agreed amount of margin collateral. In some of these cases, the transaction payments may exceed the amount of margin collateral. There shall be no advance notification of required collateral value or collateral ratios as they vary depending on the financial position of the protection buyer. CDS transactions carry the risk of losses owing to changes in the credit position of some or all of the referenced entities, and/or fluctuations of the interest rate market. The amount the protection buyer receives in the event that the CDS is triggered by a credit event may undercut the total amount of premiums that he/she has paid in the course of the transaction. Similarly, the amount the protection seller pays in the event of a credit event may exceed the total amount of premiums that he/she has received in the transaction. All other conditions being equal, the amount of premiums that the protection buyer pays and that received by the protection seller shall differ. In principle, CDS transactions will be limited to financial instruments business operators and qualified institutional investors.
Transfers of equities to another securities company via the Japan Securities Depository Center are subject to a transfer fee of up to ¥11,000 (tax included) per issue transferred depending on volume. No account fee will be charged for marketable securities or monies deposited.

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